Proving angle relationships

ANSWER KEY

Core Concepts: Proving angle relationships

1 In $\triangle ABC$ studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 94° ✓

b. 104°

c. 86°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 45^\circ) = 94^\circ$.

2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm ✓

c. 15 cm

d. 20 cm

Multiplying the original length by the scale factor gives $5 \times 2 = 10$ cm.

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 14

c. 12 ✓

d. 17

Substituting $x = 1$ into $y = 5x + 7$ gives $y = 5(1) + 7 = 12$.

Procedural Fluency & Algebraic Manipulation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 In $\triangle ABC$ studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 87° ✓

b. 97°

c. 79°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 52^\circ) = 87^\circ$.

5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm ✓

c. 20 cm

d. 45 cm

Multiplying the original length by the scale factor gives $5 \times 3 = 15$ cm.

6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 12

b. 19

c. 17 ✓

d. 22

Substituting $x = 2$ into $y = 5x + 7$ gives $y = 5(2) + 7 = 17$.

Graphical & Geometric Interpretation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 In $\triangle ABC$ studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 80° ✓

b. 90°

c. 72°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 59^\circ) = 80^\circ$.

8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm ✓

c. 25 cm

d. 80 cm

Multiplying the original length by the scale factor gives $5 \times 4 = 20$ cm.

9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?

a. 17

b. 24

c. 22 ✓

d. 27

Substituting $x = 3$ into $y = 5x + 7$ gives $y = 5(3) + 7 = 22$.

Exam-Style Problem Solving & Modeling

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 In $\triangle ABC$ studying proving angle relationships, $m\angle A = 41^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 73° ✓

b. 83°

c. 65°

d. 139°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (41^\circ + 66^\circ) = 73^\circ$.

11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm ✓

c. 30 cm

d. 125 cm

Multiplying the original length by the scale factor gives $5 \times 5 = 25$ cm.

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 22

b. 29

c. 27 ✓

d. 32

Substituting $x = 4$ into $y = 5x + 7$ gives $y = 5(4) + 7 = 27$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647